DataPreparationAndCleaning

Write your name here

2026-05-04

1. Preparations

1.1 Load the libraries

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(ez)
```

```
## Warning: package 'ez' was built under R version 4.5.2
```

1.2 Load the data

```
load("CHC9012_DataImportedFromPCIbex.RData")
```

1.3 Check the data

```
str(data)
```

```
## 'data.frame':    6195 obs. of  18 variables:
##  $ Results.reception.time             : int  1777517421 1777517421 1777517421 1777517421 1777517421
##  $ MD5.hash.of.participant.s.IP.address: chr  "965c6743ad440dd6eaf441586b84d5c2" "965c6743ad440dd6ea
##  $ Controller.name                    : chr  "PennController" "PennController" "PennController" "Pe
##  $ Order.number.of.item               : int  0 0 0 0 1 1 1 1 1 1 ...
##  $ Inner.element.number               : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ Label                              : chr  "Welcome" "Welcome" "Welcome" "Welcome" ...
##  $ Latin.Square.Group                 : chr  "NULL" "NULL" "NULL" "NULL" ...
##  $ PennElementType                    : chr  "PennController" "PennController" "PennController" "Pe
##  $ PennElementName                    : chr  "0" "0" "0" "0" ...
##  $ Parameter                          : chr  "_Trial_" "_Header_" "_Header_" "_Trial_" ...
##  $ Value                              : chr  "Start" "Start" "End" "End" ...
```

```
##  $ EventTime                        : num  1.78e+12 1.78e+12 1.78e+12 1.78e+12 1.78e+12 ...
##  $ ItemNo                           : chr  "NULL" "NULL" "NULL" "NULL" ...
##  $ Condition                        : chr  "" "" "" "" ...
##  $ Group                            : chr  "" "" "" "" ...
##  $ UserAge                          : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ UserDegree                       : chr  "" "" "" "" ...
##  $ Comments                         : chr  "" "" "" "" ...
```

2. Prepare the dataset

2.1 Remove unnecessary columns

```
data$Controller.name <- NULL
data$Inner.element.number <- NULL
data$Latin.Square.Group <- NULL
data$Comments <- NULL
```

2.2 Rename columns

```
names(data)
```

```
##  [1] "Results.reception.time"
##  [2] "MD5.hash.of.participant.s.IP.address"
##  [3] "Order.number.of.item"
##  [4] "Label"
##  [5] "PennElementType"
##  [6] "PennElementName"
##  [7] "Parameter"
##  [8] "Value"
##  [9] "EventTime"
## [10] "ItemNo"
## [11] "Condition"
## [12] "Group"
## [13] "UserAge"
## [14] "UserDegree"
```

```
names(data)[1] <- "ID_1_TimeReceived"
names(data)[2] <- "ID_2_IP"
```

2.3 Select the rows you need

```
data2 <- data %>%
  filter(Label == "Experiment")

data2 <- data2 %>%
  filter(Parameter != "_Header_")
```

2.4 Add a "Reaction time" column

```
##Calculate reaction time
RT_Prep1 <- data2 %>%
```

```
  filter(Parameter == "Choice" | Value == "Start")

RT_Prep1$Index <- ifelse(RT_Prep1$Value == "Start",
                         "Start",
                         "Choice")

RT_Prep1$PennElementType <- NULL
RT_Prep1$PennElementName <- NULL
RT_Prep1$Parameter <- NULL
RT_Prep1$Value <- NULL

RT_Prep2 <- spread(data = RT_Prep1, Index, EventTime)

RT_Prep2$Choice <- as.numeric(RT_Prep2$Choice)
RT_Prep2$Start <- as.numeric(RT_Prep2$Start)
RT_Prep2$RT <- RT_Prep2$Choice - RT_Prep2$Start

data3 <- data2 %>%
  filter(Parameter %in% "Choice")

data3 <- data3 %>%
  left_join(RT_Prep2 %>% select(ID_1_TimeReceived, ID_2_IP, ItemNo, RT),
            by = c("ID_1_TimeReceived", "ID_2_IP", "ItemNo"))
```

2.5 Change the nature of the variables

```
## Transform to "factors" (or "groups")
data3$ID_1_TimeReceived <- as.factor(data3$ID_1_TimeReceived)
data3$ID_2_IP <- as.factor(data3$ID_2_IP)
data3$ItemNo <- as.factor(data3$ItemNo)
data3$Condition <- as.factor(data3$Condition)
data3$Group <- as.factor(data3$Group)
data3$UserDegree <- as.factor(data3$UserDegree)

## Transform to numeric
data3$Order.number.of.item <- as.numeric(data3$Order.number.of.item)
data3$Value <- as.numeric(data3$Value)
data3$UserAge <- as.numeric(data3$UserAge)
data3$RT <- as.numeric(data3$RT)
```

2.6 Final removal of unnecessary columns

```
str(data3)
```

```
## 'data.frame':    1050 obs. of  15 variables:
##  $ ID_1_TimeReceived   : Factor w/ 35 levels "1777517421","1777518106",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ ID_2_IP             : Factor w/ 5 levels "500b881331cdc3f3e92d95160167b433",..: 5 5 5 5 5 5 5 5 5
##  $ Order.number.of.item: num  16 33 26 25 31 7 28 17 21 34 ...
##  $ Label               : chr  "Experiment" "Experiment" "Experiment" "Experiment" ...
##  $ PennElementType     : chr  "Scale" "Scale" "Scale" "Scale" ...
##  $ PennElementName     : chr  "Scale" "Scale" "Scale" "Scale" ...
##  $ Parameter           : chr  "Choice" "Choice" "Choice" "Choice" ...
##  $ Value               : num  6 1 4 3 2 6 1 7 1 1 ...
```

```
##  $ EventTime           : num  1.78e+12 1.78e+12 1.78e+12 1.78e+12 1.78e+12 ...
##  $ ItemNo              : Factor w/ 60 levels "1","10","11",..: 3 54 47 46 52 12 49 4 41 55 ...
##  $ Condition           : Factor w/ 2 levels "Grammatical",..: 1 2 2 2 2 1 2 1 2 2 ...
##  $ Group               : Factor w/ 2 levels "A","B": 1 1 1 1 1 1 1 1 1 1 ...
##  $ UserAge             : num  30 30 30 30 30 30 30 30 30 30 ...
##  $ UserDegree          : Factor w/ 4 levels "BA","High school",..: 4 4 4 4 4 4 4 4 4 4 ...
##  $ RT                  : num  20079 13763 13247 13997 13063 ...
```

```
data3$Label <- NULL
data3$PennElementType <- NULL
data3$PennElementName <- NULL
data3$Parameter <- NULL
data3$EventTime <- NULL

str(data3)
```

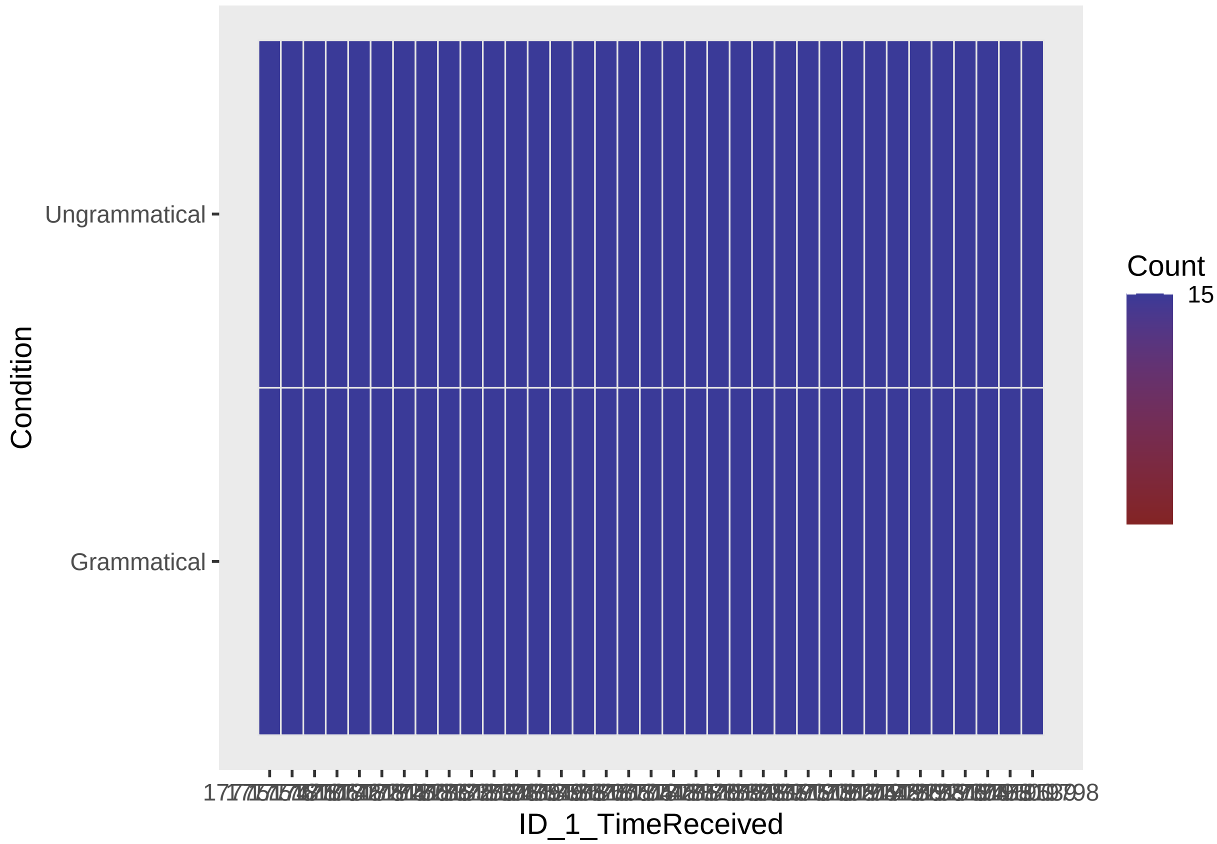
```
## 'data.frame':    1050 obs. of  10 variables:
##  $ ID_1_TimeReceived   : Factor w/ 35 levels "1777517421","1777518106",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ ID_2_IP             : Factor w/ 5 levels "500b881331cdc3f3e92d95160167b433",..: 5 5 5 5 5 5 5 5 5
##  $ Order.number.of.item: num  16 33 26 25 31 7 28 17 21 34 ...
##  $ Value               : num  6 1 4 3 2 6 1 7 1 1 ...
##  $ ItemNo              : Factor w/ 60 levels "1","10","11",..: 3 54 47 46 52 12 49 4 41 55 ...
##  $ Condition           : Factor w/ 2 levels "Grammatical",..: 1 2 2 2 2 1 2 1 2 2 ...
##  $ Group               : Factor w/ 2 levels "A","B": 1 1 1 1 1 1 1 1 1 1 ...
##  $ UserAge             : num  30 30 30 30 30 30 30 30 30 30 ...
##  $ UserDegree          : Factor w/ 4 levels "BA","High school",..: 4 4 4 4 4 4 4 4 4 4 ...
##  $ RT                  : num  20079 13763 13247 13997 13063 ...
```

2.7 Add information (The actual sentences)

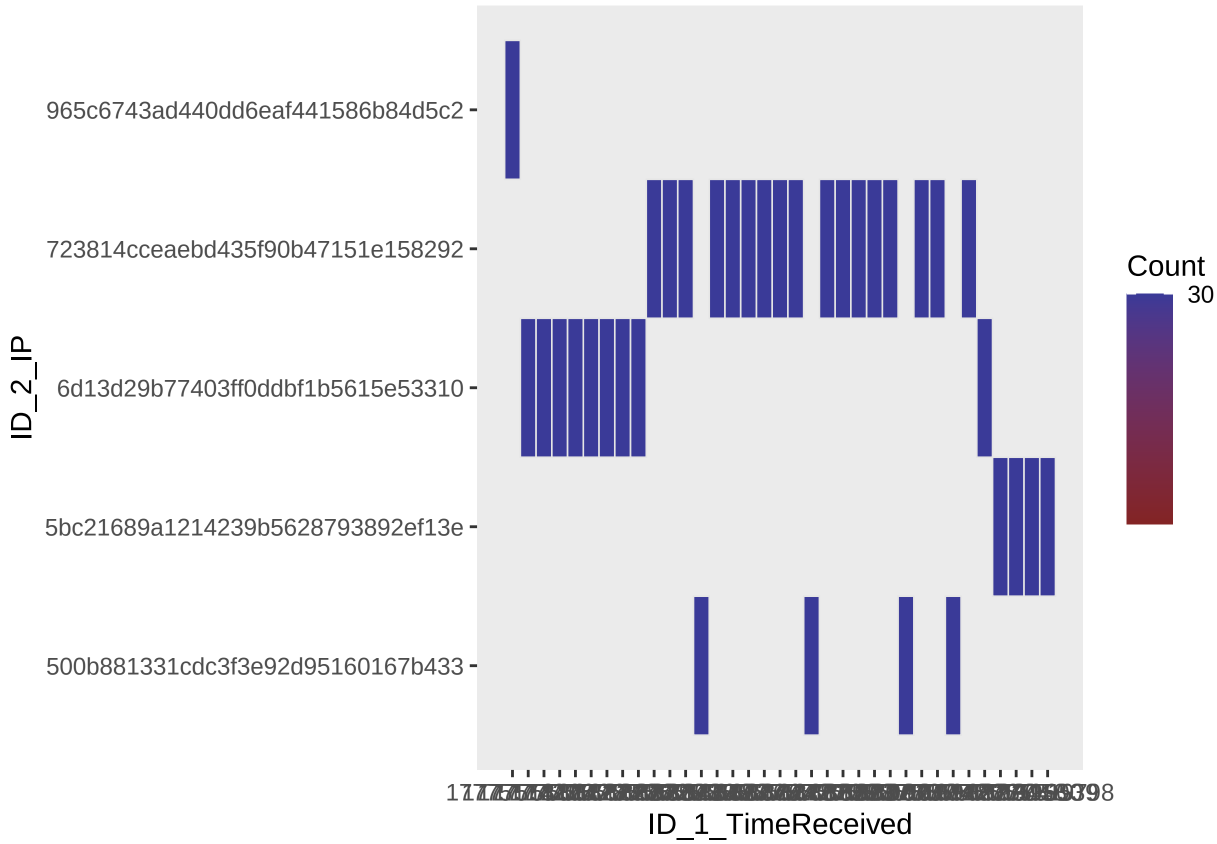
```
material <-
↪  read.csv("~/Documents/CHC9012_Spring2026/PreprocessingAnalysesR/material.csv")
material$ItemNo <- as.factor(material$ItemNo)

data3 <- data3 %>%
  left_join(material %>% select(ItemNo, Sentence),
            by = "ItemNo")
```

3. Check that everything is in order

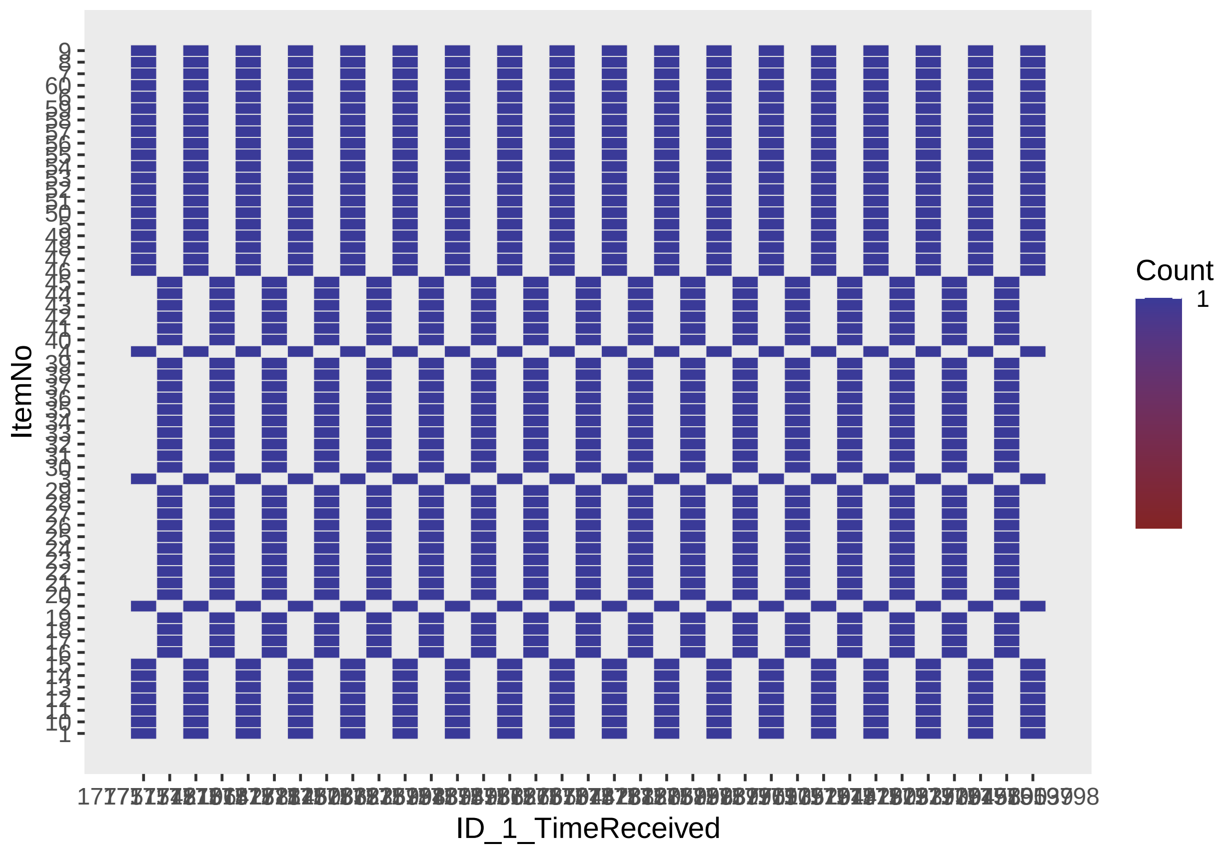
```
# Same number of item per condition for each participant? (should be '15')
ezDesign(data3, ID_1_TimeReceived, Condition)
```



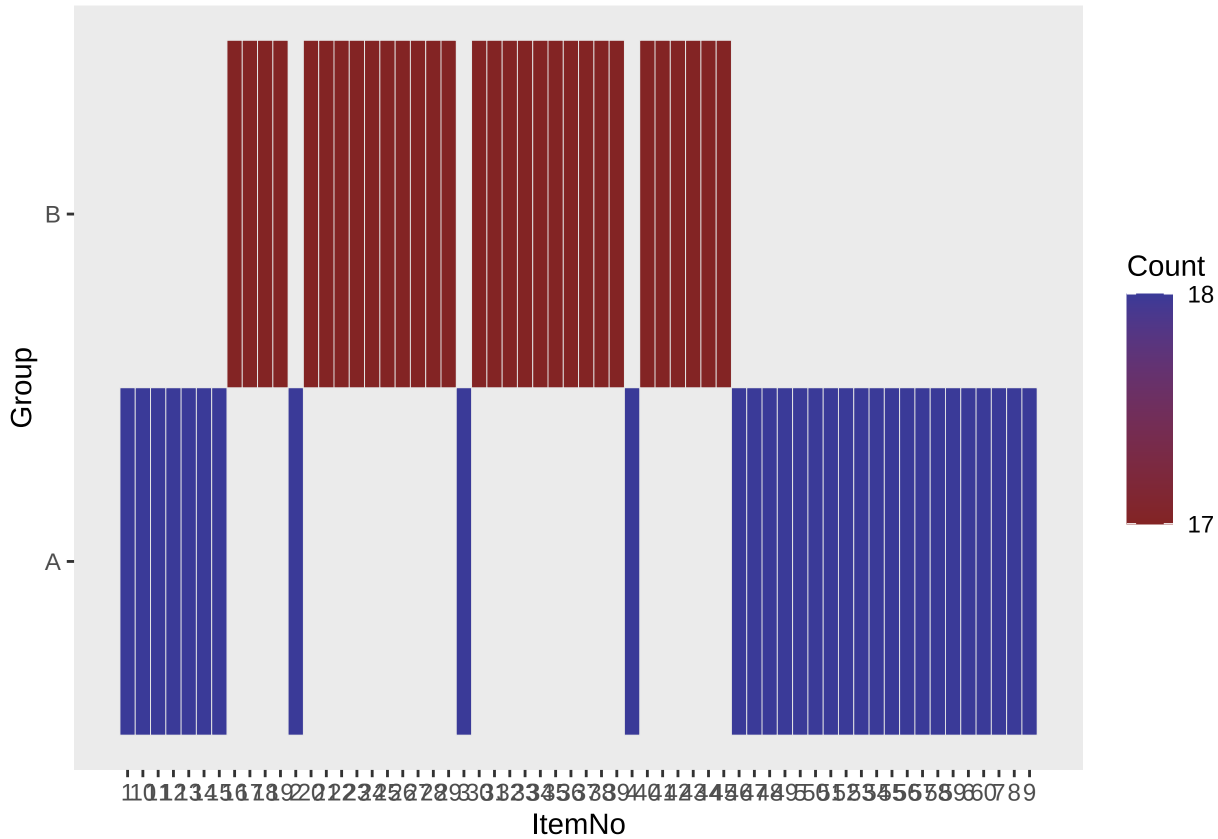

```r
# Which participants share the same ID? (hint that this could be the same participant
↪  filling the questionnaire twice)
ezDesign(data3, ID_1_TimeReceived, ID_2_IP)
```



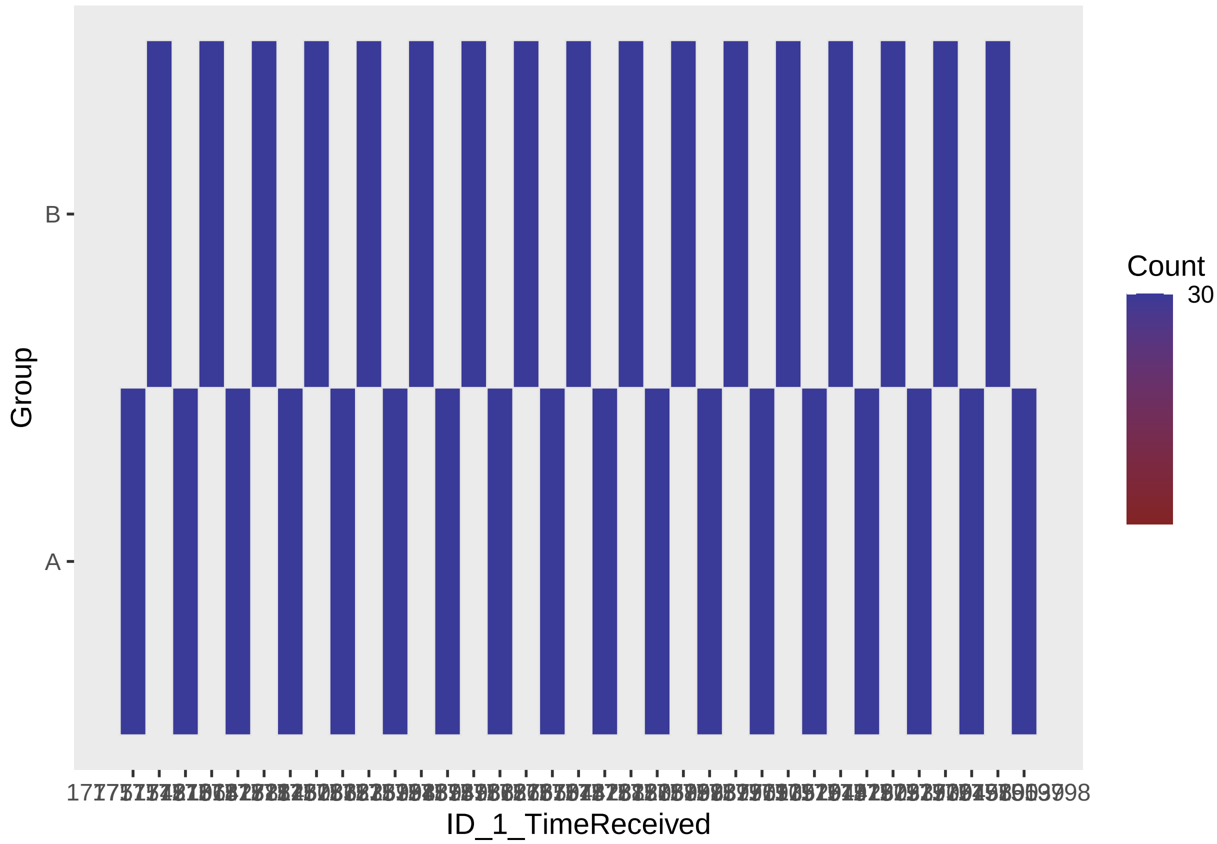

```r
# Number of times an item appeared by participant (should be '1')
ezDesign(data3, ID_1_TimeReceived, ItemNo)
```




```r
# Were the items well dispatched in the two lists?
ezDesign(data3, ItemNo, Group)
```




```
# Were the participants well dispatched in the two lists? (should be '30')
ezDesign(data3, ID_1_TimeReceived, Group)
```



4. Save the prepared dataset

```
save(data3,
     file = "CHC9012_PCIbexDataPrepared.RData")
```